QOSAIN BUKHARI

Machine Learning Engineer - NLP - CV - MLOps - GenAI / RAG

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 1+ year of freelance experience delivering production-grade ML, computer vision, and GenAI/RAG systems for direct clients, plus 8+ end-to-end projects spanning model training to Docker/CI-CD deployment. Proven ability to build and ship real-world systems: custom-trained YOLOv12 models (88.6% mAP@50), production RAG pipelines with 90% answer faithfulness, and FastAPI services backed by MLflow, DVC, and automated Pytest suites. Final-year Software Engineering student at MUST (expected graduation 2026), actively seeking a junior ML Engineer role where I can apply strong PyTorch, LangChain, and MLOps skills to solve production problems from day one.

EXPERIENCE

Freelance Machine Learning Engineer, Self-employed (Direct Clients) | *2025 - Present*

- Delivered end-to-end ML solutions for direct clients, including custom model development, chatbot systems, computer vision applications, and data pipeline engineering.
- Built and deployed machine learning models tailored to client requirements across multiple domains.
- Developed conversational AI / chatbot solutions and computer vision pipelines for real-world use cases.
- Designed and implemented data pipelines for client data processing and model-ready datasets.

TECHNICAL SKILLS

Languages: Python, SQL

ML / DL: PyTorch, Scikit-learn, TensorFlow, HuggingFace Transformers, YOLOv11/v12, OpenCV, NLTK, XGBoost

MLOps: MLflow, DVC, Docker, Docker Hub, GitHub Actions (CI/CD), Pytest, Modular Pipelines, Git

GenAI / RAG: LangChain, ChromaDB, Sentence Transformers, Groq API, Llama 3, Prompt Engineering, RAG

APIs & UI: FastAPI, Flask, Streamlit, SQLAlchemy, Alembic, JWT Authentication, Swagger / OpenAPI

Data: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, GridSearchCV

PROJECTS

Document Q&A RAG Chatbot | *2026*

github.com/QosainBukhari/Document-Q-A-RAG-Chatbot

Stack: Python, LangChain, ChromaDB, Llama 3 (Groq), Sentence Transformers, FastAPI, Streamlit, Docker, GitHub Actions

- Production RAG system: users upload PDFs, content chunked and embedded into ChromaDB, Llama 3 returns grounded answers with source citations. 85% retrieval accuracy, 90% answer faithfulness on 20 curated eval pairs.
- FastAPI backend with LangChain orchestration, persistent vector store, and multi-turn Streamlit frontend.
- Full CI/CD via GitHub Actions: automated Pytest suite covering 6 pipeline stages (load, chunk, embed, vector DB, retrieve, end-to-end) runs on every commit.

Accident Detection System - Real-Time CV + YOLOv12 (Final Year Project) | *2025-2026*

github.com/QosainBukhari/accident_detection_system

Stack: Python, YOLOv12, FastAPI, SQLAlchemy, Alembic, JWT Auth, Docker, SQLite, SMTP, Pytest, Swagger UI

- Custom-trained YOLOv12 on accident footage: 88.6% mAP@50, 87% precision, 84% recall with a low-latency inference pipeline optimised for real-time use.
- Full-stack FastAPI backend: JWT auth, SQLAlchemy ORM, Alembic migrations, real-time web dashboard, and automated SMTP email alerts on detection events.
- Production codebase: Dockerised deployment, Pytest test suite, and Swagger UI docs included.

CommerceIQ Analytics Platform - Full-Stack E-Commerce BI | *2025*

github.com/QosainBukhari/CommerceIQ_Analytics_Platform

Stack: Python, FastAPI, SQLAlchemy, Pandas, MySQL, React, Tailwind CSS, Recharts, Axios, Docker

- End-to-end analytics platform transforming raw e-commerce transaction data (Olist Brazilian dataset) into interactive BI dashboards covering revenue trends, product analytics, seller performance, geography, and payment analysis.
- ETL pipeline with Pandas into a MySQL relational database; modular FastAPI backend exposing 11 analytics endpoints (KPIs, monthly revenue/growth, top products, order status, geography) with SQLAlchemy ORM.

- React frontend with Tailwind CSS and Recharts: multi-page dashboard (7 analytics pages) with KPI cards, interactive charts, dynamic tables, and reusable components; Dockerised full-stack deployment via docker-compose.

Fake News Detection - End-to-End NLP + MLOps | *2026*
github.com/QosainBukhari/Fake_News_Detection_End_to_End_MLOPS
Stack: Python, TF-IDF, Random Forest, FastAPI, Docker, MLflow, DVC

- NLP classification pipeline (TF-IDF + Random Forest) achieving 94%+ accuracy on 40,000 samples; served via FastAPI REST API with one-command Docker Hub deployment.
- MLflow experiment tracking and DVC data versioning throughout; modular codebase with structured logging and custom exception handling.

CT-Scan Image Classification - Deep Learning + MLOps | *2025*
github.com/QosainBukhari/Chest_CT_Scan_Classification_End_to_end_Project_MLOPS
Stack: Python, PyTorch, CNN, Transfer Learning, Grad-CAM, MLflow, FastAPI, Docker

- Medical image classifier using a pretrained CNN backbone with Grad-CAM explainability heatmaps to visualise model attention, critical for clinical trust and debugging.
- Early stopping, model checkpointing, modular train/validate/infer architecture; FastAPI + Docker deployment with full MLflow experiment tracking.

Network Security Threat Detection - End-to-End ML | *2025*
github.com/QosainBukhari/End_to_end_Network_security_project
Stack: Python, Scikit-learn, Ensemble Models, FastAPI, Docker, MLflow

- Binary classification pipeline for malicious network activity: 96% AUC-ROC on held-out test set.
- Domain-specific feature engineering (packet ratios, protocol flags) improved F1 score by +8 points over baseline; deployed via FastAPI with Dockerised MLflow tracking.

EDUCATION

BS Software Engineering, Mirpur University of Science & Technology (MUST) - Mirpur, AJK
2022 - 2026 (Expected, Final Year) | CGPA: 3.35 / 4.0
Relevant Coursework: Data Structures & Algorithms, Database Systems, Statistics, Software Engineering
Final Year Project: Accident Detection System - YOLOv12 real-time pipeline, FastAPI backend, Docker, JWT Auth, automated alerting

F.Sc Pre-Engineering, Green Hills Science College Muzaffarabad (GHSC) - Muzaffarabad, AJK
2019 - 2021 | Grade: 93.7%

CERTIFICATIONS

- Complete Data Science, ML, DL & NLP Bootcamp - Udemy (Krish Naik), 2024
- Machine Learning Concepts - Microsoft, 2024
- Python for Data Science - Coursera, 2023